
MODEL SELECTION AND VALIDATION REPORT (arctic_biomass model)

OBSERVATIONS:1023

SANITY: OK - all models in the selection process passed sanity checks.

SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

MODEL SUMMARY (arctic_biomass model)

Spatiotemporal model fit by REML ['sdmTMB']

Formula: fixed_formula

Time column: year

Family: lognormal(link = 'log')

Conditional model:

	coef.est	coef.se
(Intercept)	7.03	0.96
kde_fric	0.25	0.04
kde_feve	-0.30	0.03
tric	-0.15	0.04
teve	-0.42	0.04
sbt_mean	0.12	0.04
sic_mean	0.01	0.07
sbt_sd	0.10	0.06
sst_sd	0.06	0.05
log_chla_mean	-0.01	0.04
depth	0.11	0.05

Dispersion parameter: 0.82

Spatiotemporal AR1 correlation (rho): 0.97

Matérn range: 891.68

Spatiotemporal marginal AR1 SD: 0.97

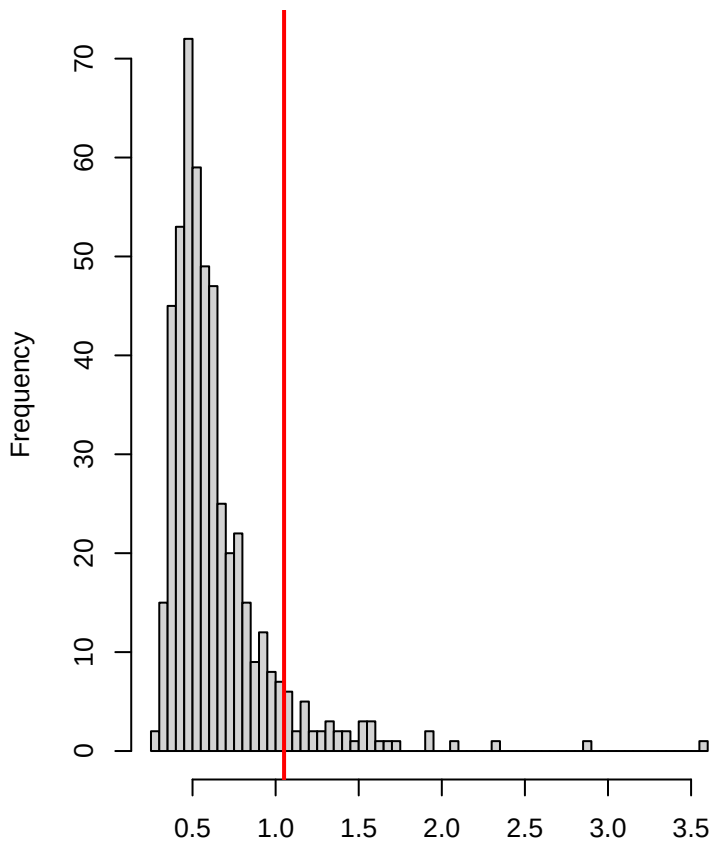
REML criterion at convergence: 8284.778

See ?tidy.sdmTMB to extract these values as a data frame.

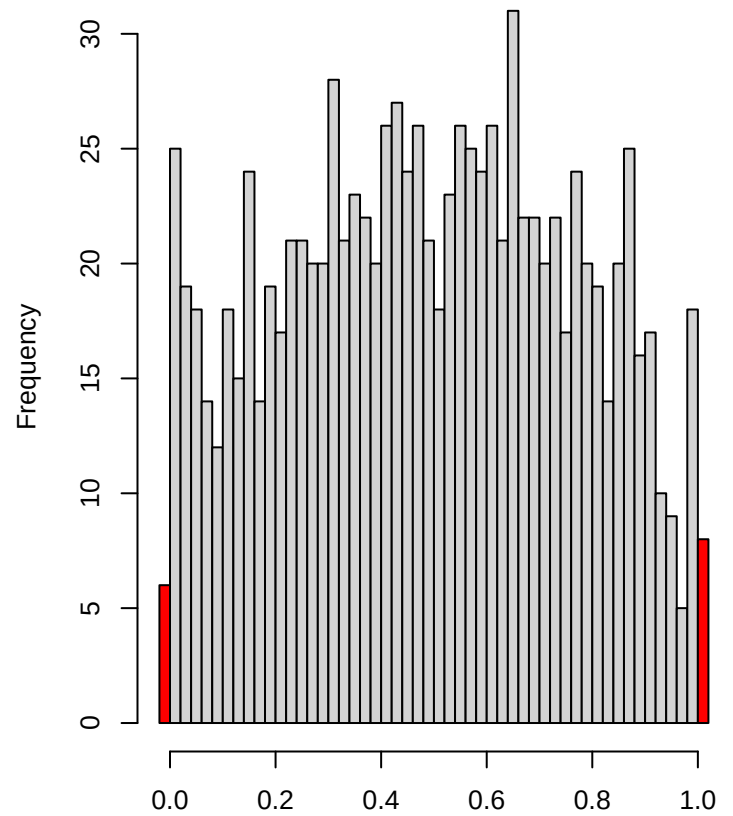
SELECTION STATISTICS

	df	AIC	delta_AIC
spatiotemporal_random_fields_ar1_with_anisotropy	17	16598.70	0.8559119
spatiotemporal_random_fields_ar1	15	16599.56	49.8191440
spatial_random_fields	14	16649.38	1.3889371
spatial_random_fields_with_anisotropy	16	16650.76	5.4705314
spatiotemporal_random_fields_iid_with_anisotropy	16	16656.23	12.7625288
spatiotemporal_random_fields_iid	14	16669.00	82.3513202
error_only	12	16751.35	NA

DHARMA nonparametric dispersion test via sd of residuals fitted vs. simulated



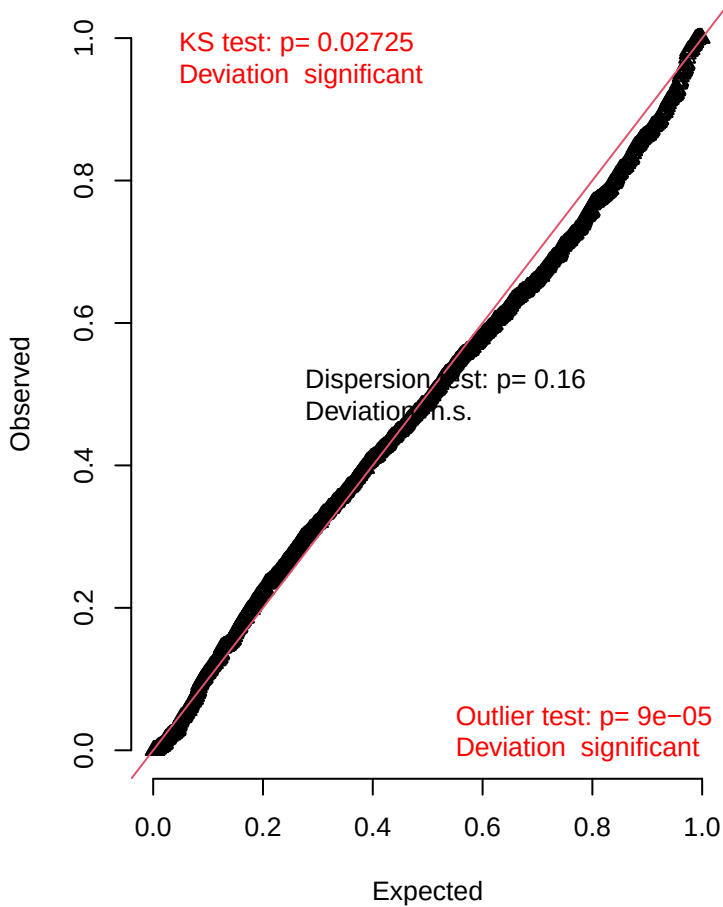
Outlier test significant



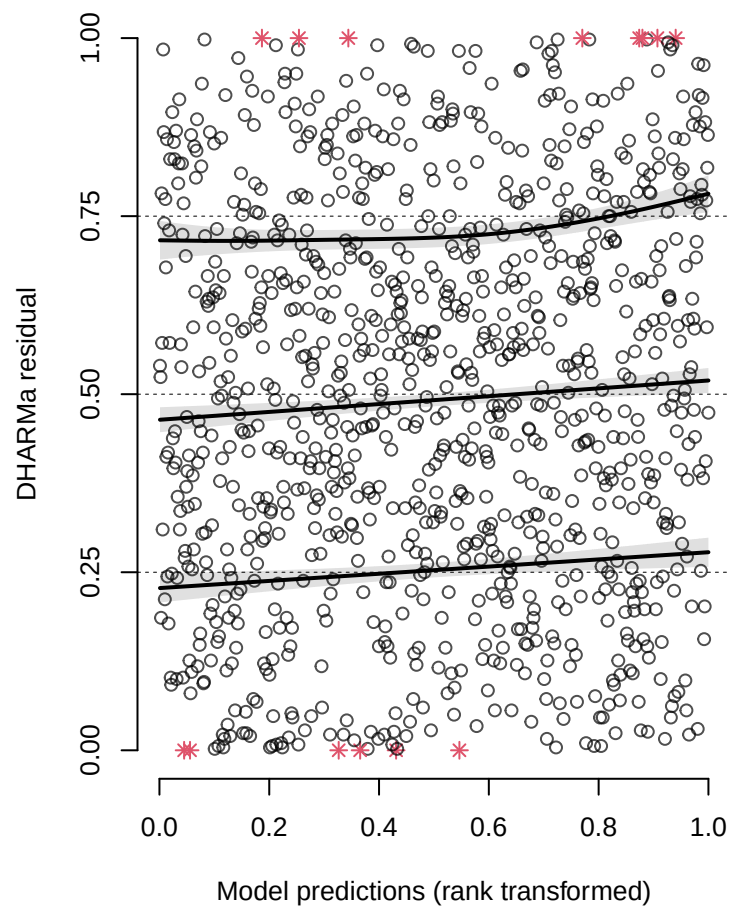
Simulated values, red line = fitted model. p-value (two.sided) :

Residuals (outliers are marked red)

QQ plot residuals



DHARMA residual vs. predicted
No significant problems detected



DHARMa Moran's I test for distance-based autocorrelation

Moran's I = -0.002 | H1: Distance-based autocorrelation | p-value = 0.905

